

EE4C5/5C5 Digital Signal Processing.

Lecture 1 – An Introduction.

Signal Processing?

- ... is concerned with representation, transformation and manipulation of signals and the information they contain.
 - (Oppenheim and Schafer, module textbook)

Digital?

- Discrete-time?
 - What's the difference?
- Analog signals in nature?
 - Electrical, acoustic, mechanical? (name some specific ones?)
- Analog Processing?
 - Linear: amplify, filter, integrate, differentiate.
 - Non-linear: Square, rectify, invert.
- Implementation
 - Circuits, mechanical devices.

Limits of analog processing

- Accuracy impacted by operating conditions (temperature, component over time).
- Dynamic range restricted.
- Noise sensitivity.
- Inflexible.
- Data storage and transmission.
- Limited speed.

Advantages of Digital Processing

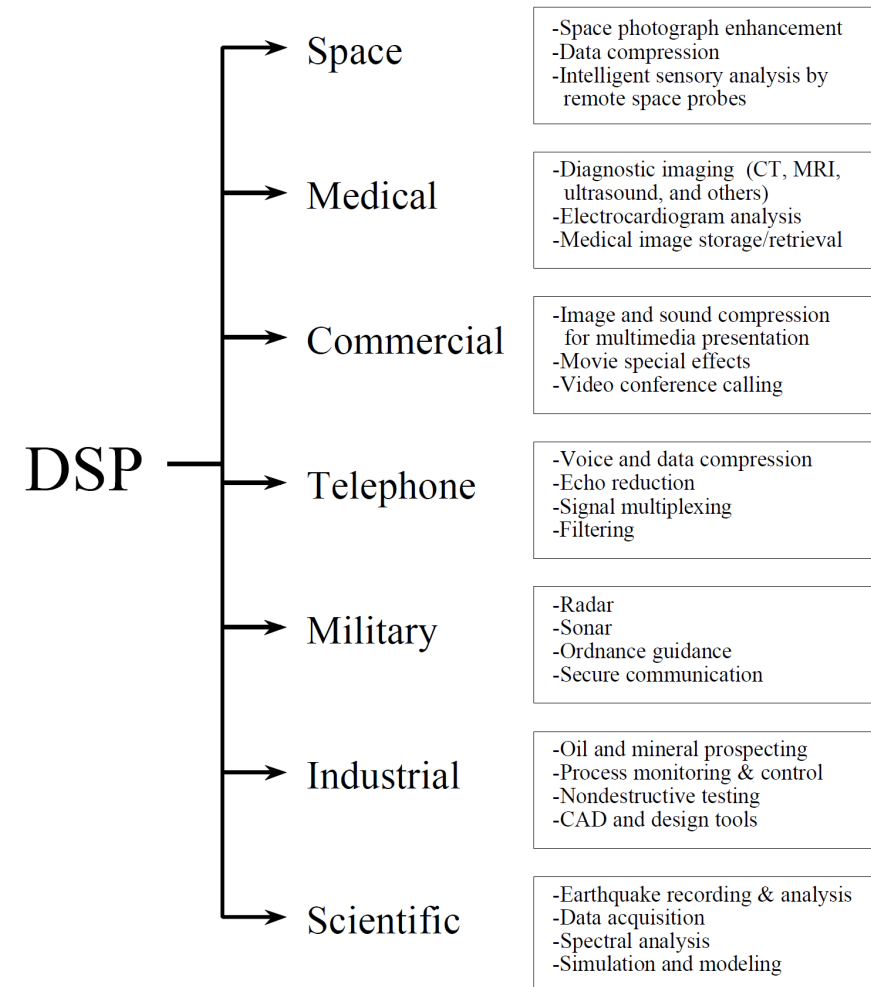
- Storage and transmission.
- Flexible.
- Can be more complex.
- Efficient implementation.
- Speed increases with technology.
- High accuracy.
- Dynamic range.
- Parallel processing.

How?

- Analog signals converted to digital (binary sequences).
- Implementation of DSP operations.
 - FPGA, ASIC, DSP Processor, CPU, GPU.

Typical Applications

- Source of diagram: Chapter 1
“*The Scientist and Engineer's Guide to Digital Signal Processing*”, Steven Smith.
- <https://www.dspguide.com/>
- Read Chapter 1.



My interest in DSP

- Professor in Speech Technology.
- Speech fundamental in human communication.
- Speech science and speech technology.
 - Audio-visual speech recognition.
 - Multimodal conversation analysis.
 - Speech synthesis evaluation.
 - Speech quality evaluation.
- Wider applications
 - Capture, compression, encoding, transmission, reconstruction, recognition, synthesis.....

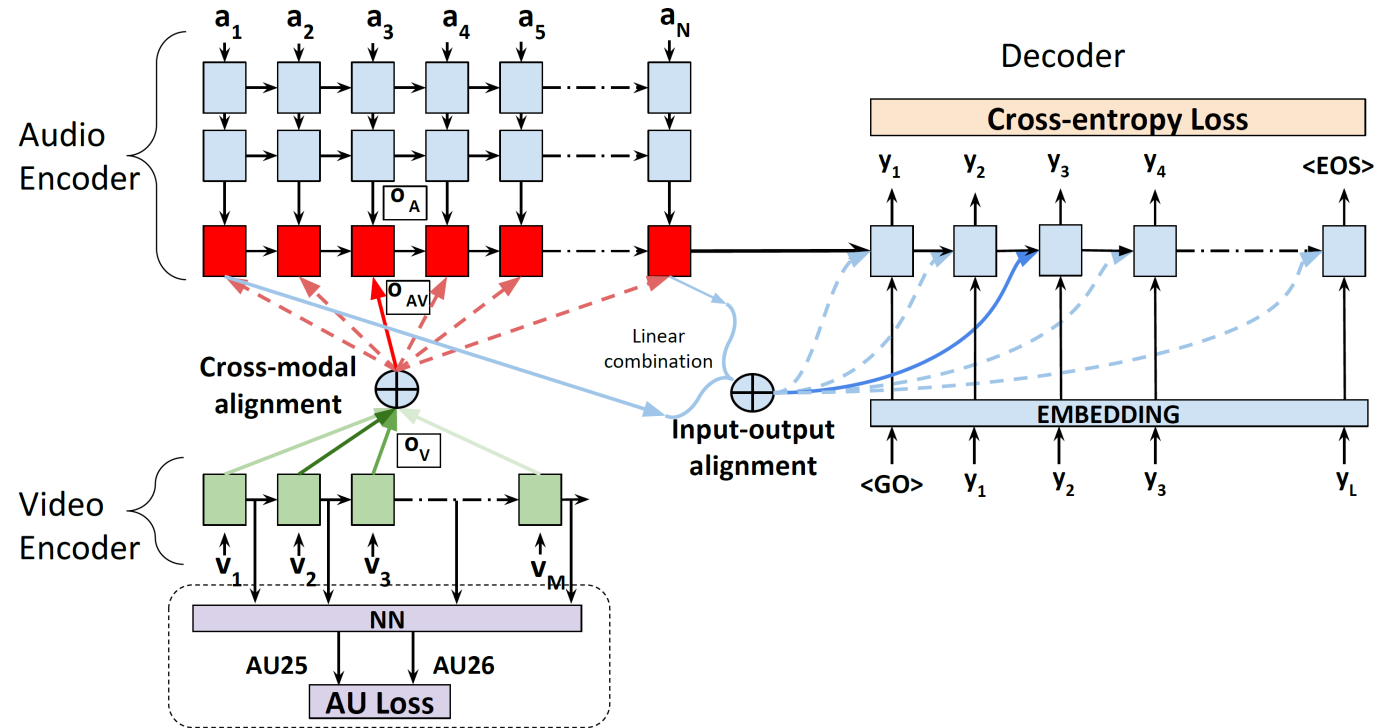
Speech signal processing



Victor Zue, MIT, <https://www.youtube.com/watch?v=j-fgbfi0W34&t=3738s>

Audio-Visual Speech Recognition

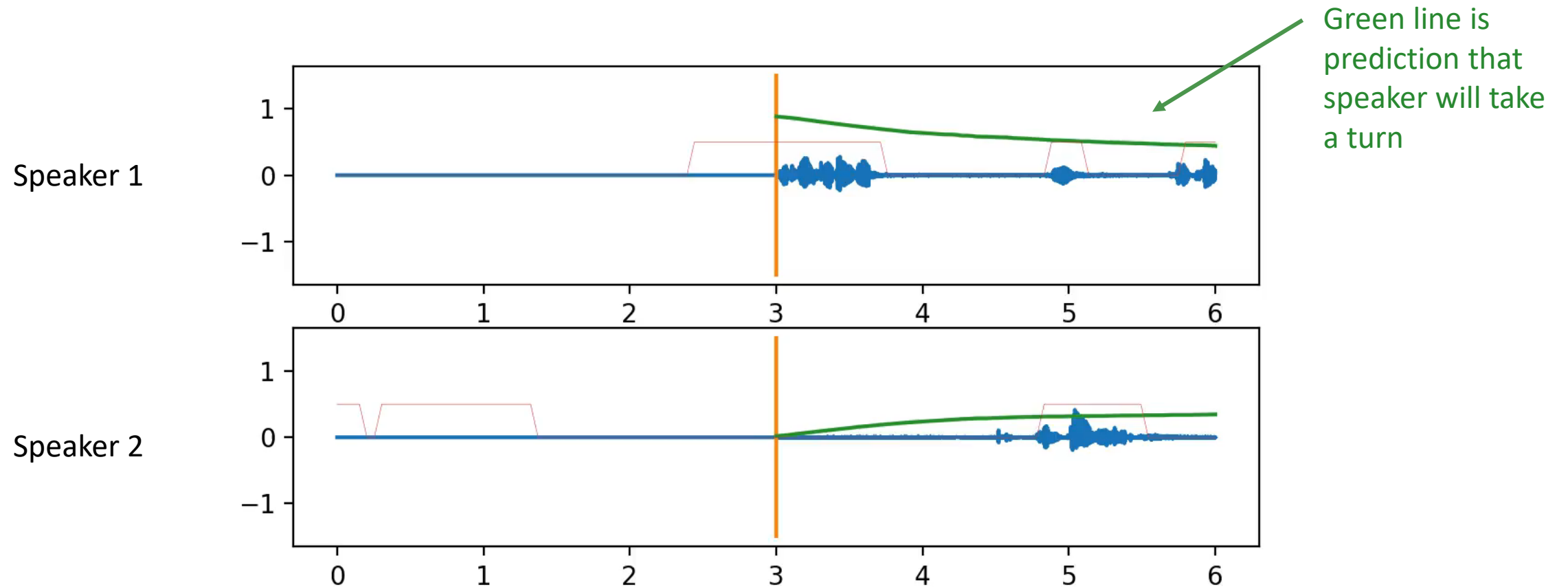
- The top layer cells of the Audio Encoder take audio representations from a stack of LSTM layers (o_A) as inputs and attend to the top layer outputs of the Video Encoder (o_V , only one layer shown), producing the crossmodal alignment.
- The Decoder receives the fused Audio-Visual representations (o_{AV}), producing an input-output alignment through a second attention mechanism.



Sterpu, George, Christian Saam, and Naomi Harte. "How to teach DNNs to pay attention to the visual modality in speech recognition." *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 28 (2020): 1052-1064.

<https://github.com/georgesterpu/avsr-tf1>

Turn taking prediction in action



Following Conversations in Noise

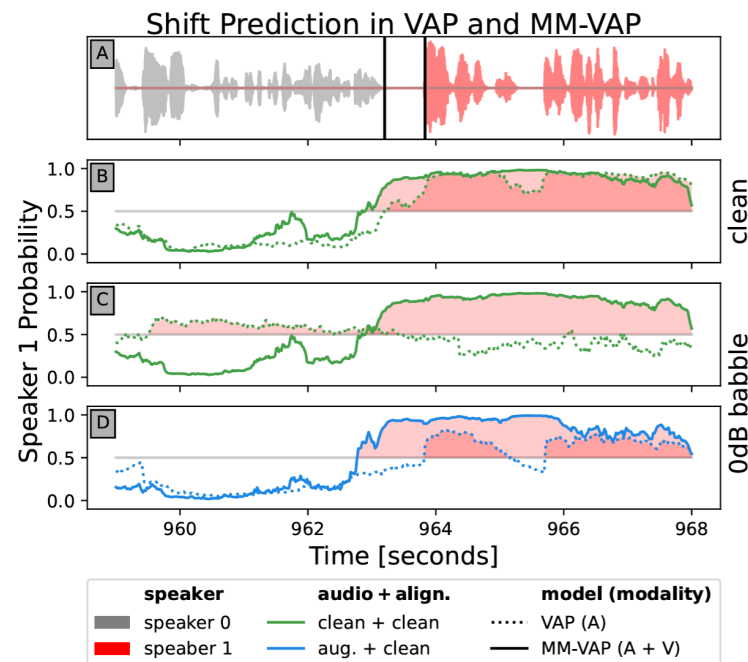


Figure 3: Model output during a shift in the Candor corpus (unseen during training). Speaker 0 (grey): "So uhm would you feel comfortable telling me about what you do for a living?", Speaker 1 (red) "Uhm yeah well for the most part I test and tweak algorithms". VAP and MM-VAP model output is shown in clean speech (B) and 0 dB babble speech (C) and (D).

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17-21 August 2025, Rotterdam, The Netherlands



Visual Cues Support Robust Turn-taking Prediction in Noise

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Abstract

Accurate predictive turn-taking models (PTTMs) are essential for naturalistic human-robot interaction. However, little is known about their performance in noise. This study therefore explores PTM performance in types of noise likely to be encountered once deployed. Our analyses reveal PTTMs are highly sensitive to noise. Hold/shift accuracy drops from 84% in clean speech to just 52% in 10 dB music noise. Training with noisy data enables a multimodal PTM, which includes visual features to better exploit visual cues, with 72% accuracy in 10 dB music noise. The multimodal PTM outperforms the audio-only PTM across all noise types and SNRs, highlighting its ability to exploit visual cues; however, this does not always generalise to new types of noise. Analysis also reveals that successful training relies on accurate transcription, limiting the use of ASR-derived transcriptions to clean conditions. We make code publicly available for future research.

Index Terms: multimodal interaction, turn-taking, human-robot interaction, noise

This paper asks if multimodality confers similar performance benefits to PTTMs in noisy environments. Surprisingly, PTTMs have to date only been tested on interactions free from background noise interference [4, 11]. The impact of noise on PTM performance is therefore unknown and unquantified. This presents a major oversight as PTTMs will be deployed *in-the-wild*, where they will inevitably encounter diverse sources and levels of noise. We therefore present the first exploration of PTM performance in background noise, asking:

1. How is the performance of predictive turn-taking models affected by background noise? and
2. Does the inclusion of visual features make multimodal PTTMs robust to background noise?

This paper explores the crucial capability of a PTM to distinguish holds (no speaker change) from shifts (changes between speakers). In noise-free conditions, a multimodal PTM which incorporates visual features outperforms an audio-only model (84% and 80% accuracy, Table 1). When we add artificial music, babble, and speech noise, the accuracy of both PTTMs is poor (51% and 52% in music). Including noise during training increases accuracy (61% and 72% in music). It also

1. Introduction

Synthetic Speech – guess what's what?



- Natural
- HMM Synthesis (I)
- DNN + WORLD
- DNN + NSF

- Hybrid System (M)
- FastPitch + WaveRNN
- FastPitch + WaveNet

Required Reading & other material

- Chapter 1 “The Scientist and Engineer's Guide to Digital Signal Processing”, Steven Smith.
- Discrete-Time Signal Processing. Oppenheim & Schaffer, Chapter 1, Intro. (O&S in future notes)
- Introduction to Matlab.
 - <https://matlabacademy.mathworks.com/details/matlab-onramp/gettingstarted> (2-hour intro if your Matlab is rusty)
- Future of Signal Processing.
 - <https://futureofsp.eecs.mit.edu/videos/>

Deepfakes

- <https://www.youtube.com/shorts/-cROWwDtinA>

The 4C5/5C5 Module

4C5/5C5

- Module has two codes
- 4C5
 - 4th years and any undergraduates
 - Pass 40%
- 5C5
 - 5th year MAI, MSc, and PhD students
 - Pass 50%
- Make sure you are enrolled in the correct module
 - Mostly applies to visiting students

Course Content

- Signals and Properties, LTI systems – review (3C1 or equivalent background).
- Sampling and reconstruction, Decimation, interpolation, quantisation.
- Digital Filters, Linear Phase systems.
- Filter design methods – IIR, FIR.
- Optimum FIR Filter Design.
- Practical Filter Design in Matlab.
- Discrete Fourier Series, Discrete Fourier Transform, Discrete Fourier Transform Properties.
- Computation of FFT.
- Random Processes.
- Spectral analysis of signals with DFT.
- Filter realisations.
- DSP in an era of Deep Learning.

Module Descriptor

- Online, Engineering School website.
- Learning Outcomes.
 - LO1. Outline and use a variety of approaches to sampling and reconstruction of signals .
 - LO2. Describe, appraise and implement filter design methods for IIR and FIR filters, identifying trade-offs and evaluating outcomes.
 - LO3. Appreciate and illustrate the role of linear phase.
 - LO4. Elaborate on the relationship between the Continuous Time Fourier Transform and the Discrete Fourier Transform.
 - LO5. Discuss the importance and relevance of properties of the DFT.
 - LO6. Illustrate fast algorithms for implementation of the DFT and their practical use and advantages.
 - LO7. Interpret and analyse signals using spectral analysis techniques derived from the DFT.
 - LO8. Appreciate the role of signal processing in current approaches to system design.
 - LO9. Exploit assigned reading and lab exercises to deepen insights into module content.

5 ECTS Course

- 1 ECTS = 20-25 hours student effort.
- 4C5 – 100-125 hours.
 - 36 hours lecture.
 - 4X 1-hour lab clinics.
 - 4 Problem Solving Slots
 - Rest from you!

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44 hours

56-81 hours???

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44 hours to attend

- 3 X prep lab at home of 4hr (12).
- 1 hour study per hour of lecture (30).
- 1 hour per tutorial/homework (7).
- 4 hours prep for in-person Matlab Assignment
- 12 hours for intense exam prep.



61 hours

Lectures

- Notes shared on Blackboard.
- Sparse.
- Take notes!
- Lecture Slots:
 - All 4 slots for first few weeks.
 - Video solutions of Tutorials provided, plus Office Hours
 - Will assign Problem Solving Sessions later in Semester
- LENS Accommodations
 - Talk to me directly before you record lectures please.

Tutorials

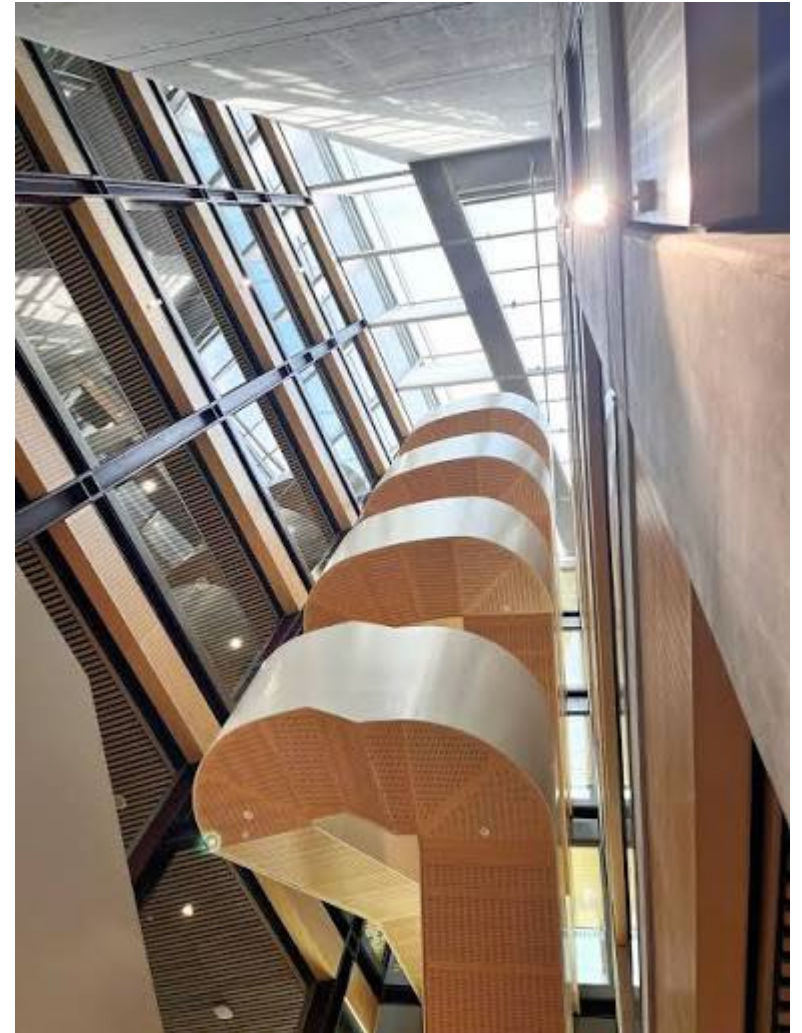
- Will assign problems 1 week in advance.
 - Sometimes from O&S textbook
- Traditional weekly Tutorial slot?
 - Class so large.
 - Not convinced it is useful for students.
- Provide video solutions.
 - Student feedback very positive last two years
 - Plus office hours (i.e. you come to my office if you have a question)
- Will hold 4 problem solving sessions as “super tutorials” if there’s demand

Lab Preparation

- How familiar are you with MATLAB?
- Quiz and Lab 0 on Blackboard.
 - Lab content area.
- Do the quiz.
 - Matlab proficiency sanity check.
 - If you can't score higher than 80% (without help!!) then do Lab 0.
- Tues of Week 2
 - Optional clinic for questions around Lab 0.
 - Expect less than half of you to need it.

Lab Clinics – start in Week 3

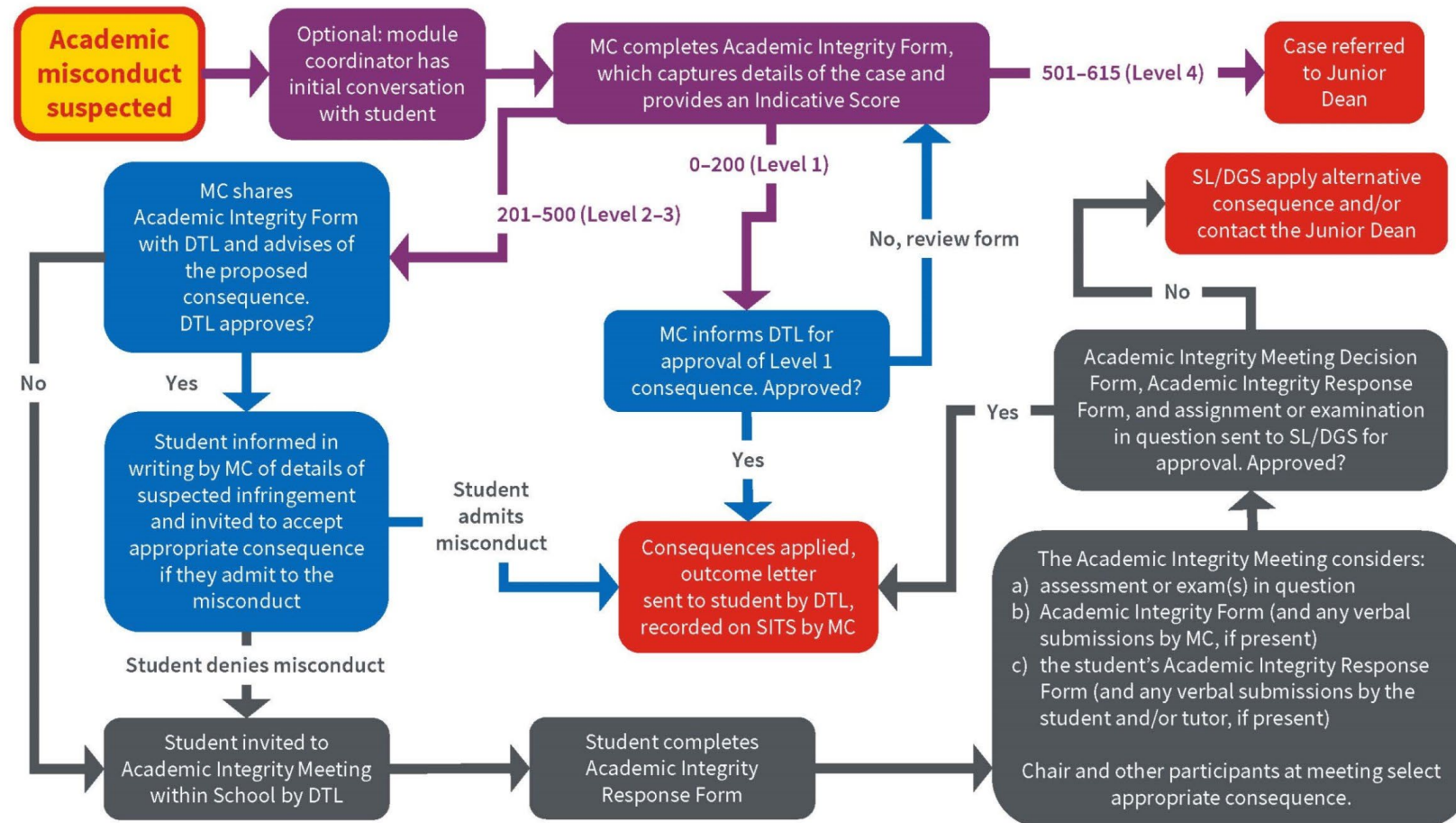
- 3 labs – A, B, C.
 - Lab A will be shared in Week 2
- Largely self-guided.
- Will get one 1-hour slot in DML as clinic per lab:
 - Tuesday afternoon.
- You do the lab BEFORE you attend clinic.
- Submit required evidence via Blackboard.
 - Not graded, formative
- Attendance at clinic optional but gets recorded.
- More on submission requirements later



Academic Integrity

- Plagiarism is the act of presenting the work or ideas of others as your own without due acknowledgement.
- Individual assignments.
- Difference between helping each other and plagiarising the work.
- Updated College system.
- Get informed:
 - <https://www.tcd.ie/teaching-learning/academic-affairs/academic-integrity/academic-misconduct-procedure/>
 - <https://www.tcd.ie/teaching-learning/academic-affairs/academic-integrity/>

Academic Misconduct



GenAI use in this module

- Use of tools like ChatGPT/Claude to write code or any part of a submission is NOT permitted in this module.
- Can be a useful study tool.
- Be wary of using it to answer past exam questions and learning those off:
 - Not well suited to this module as answers test insight and not just information reproduction.
 - Very vague and will score poorly.
- More details when we get to the labs.

Marking

- 85% End of year Exam.
- 15% CA.
- Note supplemental based 100% on exam.

What are the CA marks for?

- Lab A B and C will act as preparation material
 - You submit your code as a record
- Were marked in previous years.
 - You are all using GenAI so there is no point TAs spending hours correcting this material
- Labs are formative
 - Get feedback and support at clinics
- Will be an end of year in-person Lab
 - Matlab based
 - Use the code you previously submitted and your learning from previous lab work.
 - Do in person, submit code, a few screengrabs, answer a few questions.

Schedule (subject to change)

Teaching Week	Labs associated tasks for you	Scheduled DML Slots
Week 1	Matlab Quiz & Lab 0 if needed	-
Week 2	Lab A – do in own time	Lab 0 Clinic, optional attendance
Week 3	Lab A – do in own time	-
Week 4		Lab A Clinic in DML, optional attendance
Week 5	Lab B – do in own time	-
Week 6	Lab B – do in own time	-
Week 7 (Reading Week)		-
Week 8	Lab C – do in own time	Lab B Clinic in DML, Optional attendance
Week 9	Lab C – do in own time	-
Week 10	Have completed ALL labs	Lab C Clinic in DML, Optional attendance
Week 11		CA in-person individual assignment (Test)
Week 12		CA in-person individual assignment (Test)

} One of these slots

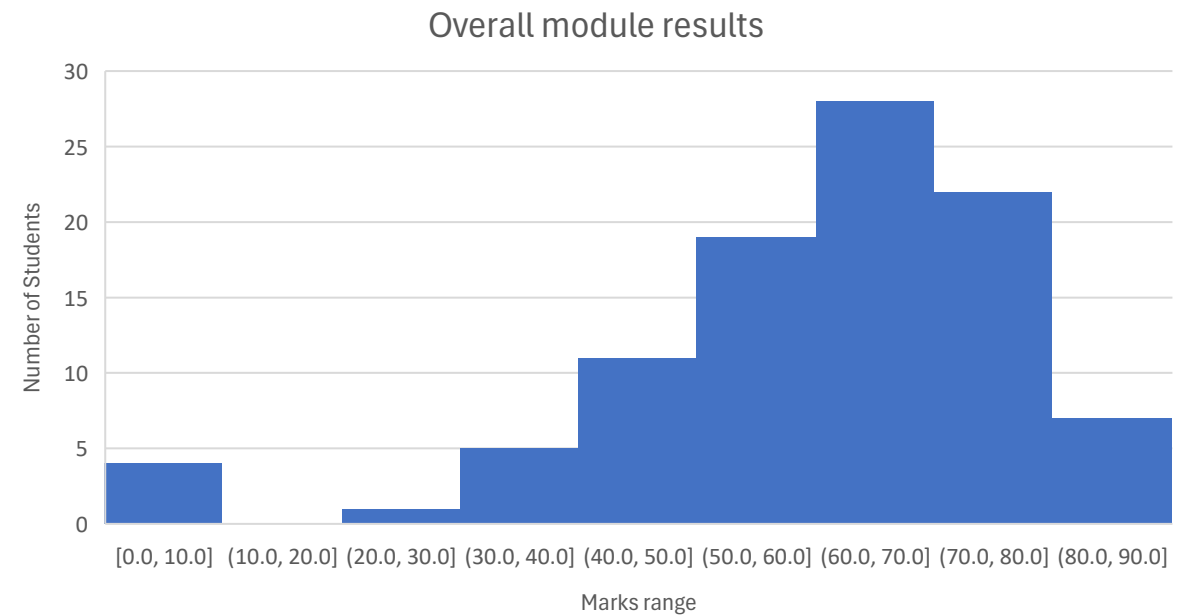
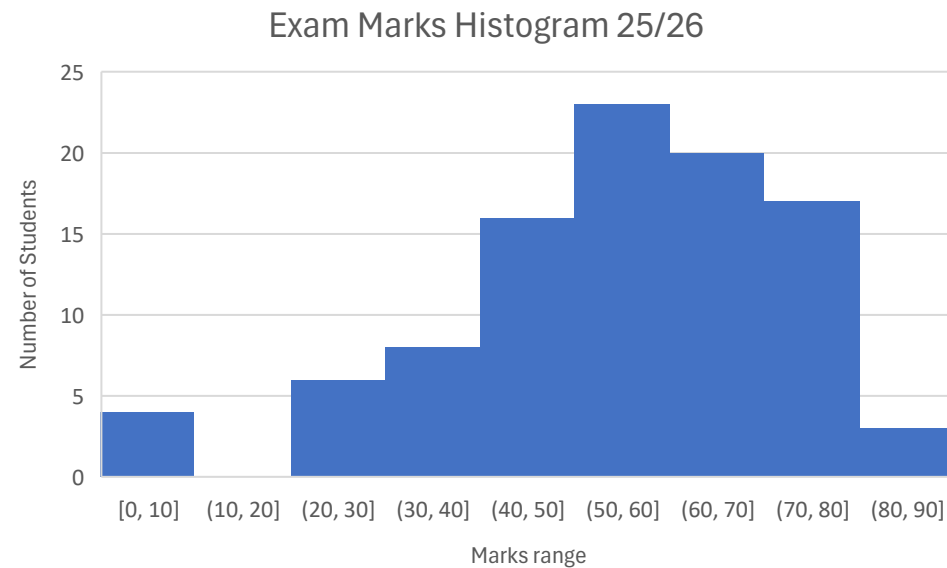
Class test

- Matlab task similar to what you have done in the assignments
- Use code you wrote before (will check it matches)
- Attendance is compulsory
 - Miss it and get zero marks
 - Think of it like end of semester exams

How to pass this module

- Attend the lectures
- Take notes at lectures
 - Many worked examples
 - Explanatory diagrams etc
- Read your notes from previous lecture before the next one
- Use the book to explain concepts you still don't understand
- Do the tutorials without the solutions in front of you
- Use the tutorial solutions to check your work
- Attend office hours for help
- Use the Lab Clinics and ask the TAs questions when there

Marks from 25/26

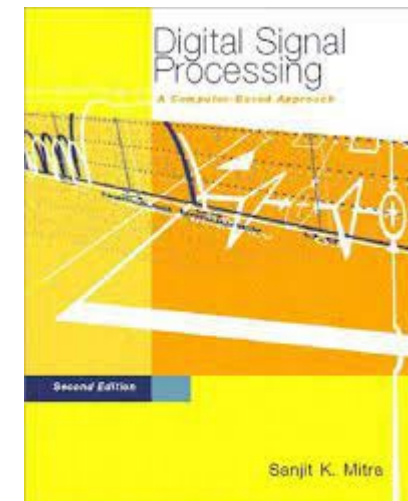
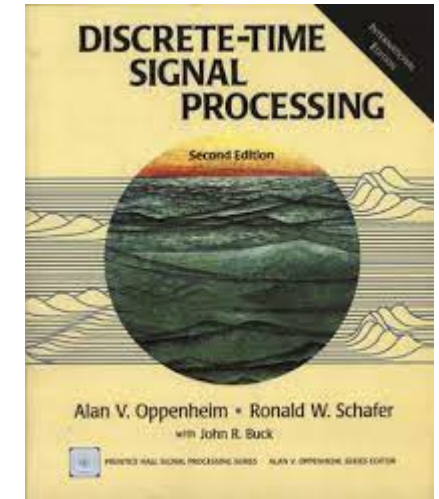
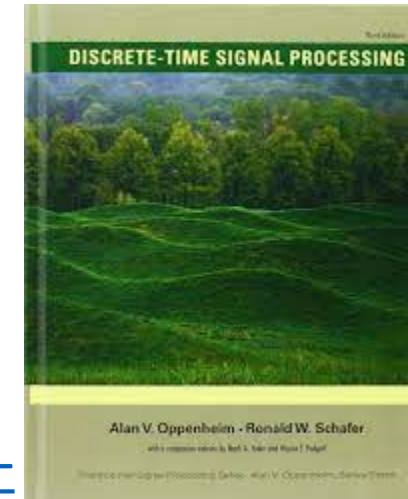


Announcements

- Module run via Blackboard.
- Check there for announcements in between classes.
- Set your email preferences.
- Other announcements will be made in lectures.
 - (I assume you are here 😊)

Module Texts

- Discrete-Time Signal Processing, Alan V. Oppenheim, Ronald W. Schafer (3rd edition).
- Online via library, or 7 S-LEN copies.
- [The Library of Trinity College Dublin: Stella Search - Discrete-time signal processing / Alan V. Oppenheim, Ronald W. Schafer. \(tcd.ie\)](#)
- Other material:
 - The Scientist and Engineer's Guide to Digital Signal Processing, Steven W. Smith, Ph.D. (free online).
 - Digital Signal Processing: A Computer-based Approach, Sanjit K. Mitra (out of print).
 - <https://ocw.mit.edu/courses/res-6-008-digital-signal-processing-spring-2011/pages/introduction/> (videos from Alan Oppenheim)



Contacting me

- Happy to take questions...
 - During class.
 - TA at lab clinics.
- Happy to take feedback.
 - Class reps.
- Please email me at:
 - nharte@tcd.ie
 - Always put 4C5 or 5C5 in the subject line as I filter emails.
 - Ensure email comes from your @tcd.ie email address.
 - I won't respond to non-college email.
 - Allow 2-3 working days for a response!